Smart Health Monitoring System Using an STM32 (ARM Cortex-M4) Microcontroller

Project Report · October 2026

Real-time vital-sign monitoring with EMA filtering, multi-timeframe trend analysis, pattern detection and risk-based alerting

Abstract

This report describes the design and implementation of a portable, real-time health monitoring system built around an STM32F401RE microcontroller (ARM Cortex-M4). The system measures three physiological parameters: body temperature (DS18B20 digital sensor), heart rate (analog pulse sensor on the 12-bit ADC) and respiratory rate (digital sound sensor). Raw readings are smoothed with an Exponential Moving Average (EMA) filter, stored in fixed-size circular buffers and analysed over three timeframes (5 seconds, 15 seconds and 1 minute). Spike, drop and oscillation patterns are detected statistically, and a Gaussian-normalised composite risk model converts all of this into a single health score between 0 and 100. A five-level, severity-based alert scheme drives a 16×2 I2C LCD and a PWM buzzer, while a UART link streams detailed logs to a PC. The entire pipeline runs on a resource-constrained microcontroller with fixed memory usage, demonstrating that useful trend-aware health analytics are feasible on low-cost embedded hardware.

1. Introduction

1.1 Background

Healthcare is steadily moving from episodic clinical measurement towards continuous, portable monitoring. An ageing population, the growth of remote patient care and the need for early detection of deterioration all increase demand for compact devices that can observe vital signs outside hospital settings. A single reading of temperature or pulse says little on its own; the direction and speed of change, and the presence of abnormal patterns, often carry more diagnostic value than any one number.

1.2 Problem Statement

Low-cost sensors are noisy, and simple threshold alarms either miss slow deterioration or raise false alarms on momentary disturbances. The challenge addressed here is to build an embedded system that (a) acquires several vital signs, (b) removes noise without losing responsiveness, (c) recognises trends and abnormal patterns, and (d) summarises overall condition as an interpretable score with graded alerts, all within the limited memory and processing budget of a microcontroller.

1.3 Scope

The project covers sensor interfacing, firmware design, signal-processing and scoring algorithms, user-interface output (LCD, buzzer, serial log) and a working hardware prototype. It is an educational and research prototype and is not a certified medical device.

2. Objectives

1. **Real-time physiological monitoring:** continuously measure body temperature, heart rate and respiratory rate.
2. **Signal conditioning:** reduce sensor noise with EMA filtering while keeping the response sensitive to genuine change.
3. **Multi-timeframe trend analysis:** track short-term and longer-term behaviour at 5-second, 15-second and 1-minute intervals.
4. **Pattern detection:** identify spikes, drops and oscillations that may indicate instability.
5. **Risk scoring:** fuse all parameters into a single 0–100 health score using Gaussian normalisation.
6. **Smart alerting:** provide severity-based audio-visual alerts proportional to the computed risk.
7. **Data logging:** stream readings, trends and analysis results over UART for documentation and offline review.
8. **Efficient implementation:** keep memory usage fixed and small using circular buffers.

3. System Overview

3.1 Architecture

The system follows a modular, layered architecture centred on the STM32F401RE. A sensor layer acquires raw data, a data-processing layer filters and analyses it, a decision layer produces the health score and alert level, and an output layer presents the result.

Component	Function	Interface	Purpose
STM32F401RE	Main controller	–	Central processing and control
DS18B20	Temperature sensor	1-Wire (GPIO)	Body temperature monitoring
Pulse sensor	Heart-rate sensor	ADC (analog)	Cardiovascular assessment
Sound sensor	Breathing sensor	Digital GPIO input	Respiratory-rate estimation
16×2 LCD	Visual output	I2C	Real-time readings, trends, score and alerts
Buzzer	Audio alert	PWM (TIM2)	Graded audible notification
ST-LINK virtual COM port	Data logging	UART (USART2)	Serial log to a PC terminal

3.2 Processing Pipeline

Stage	Operation	Description
1. Acquisition	Sensor sampling	Temperature, pulse and breathing events are read every 5 seconds
2. Pre-processing	EMA filter	Smooths each parameter (α = 0.3)
3. Storage	Circular buffers	36 samples (3 minutes) per parameter
4. Analysis	Trend analysis	Short-term average, rate of change and direction; long-term statistics
5. Analysis	Pattern detection	Spike, drop and oscillation recognition

Stage	Operation	Description
6. Evaluation	Health scoring	Gaussian risk model with rate and pattern terms, 0–100
7. Output	Alerts and logs	LCD message, buzzer pattern and UART report

3.3 Key Features

- **Multi-modal sensing** of temperature, cardiac and respiratory activity in one device.
- **Adaptive EMA filtering** with a configurable smoothing factor.
- **Circular-buffer storage** giving constant memory use regardless of run time.
- **Multi-timeframe analytics** (5 s, 15 s, 1 min) for early detection of gradual or abrupt change.
- **Statistical anomaly and pattern detection** against a rolling baseline.
- **Gaussian health scoring** that gives smooth, bounded and interpretable risk values.
- **Five-level alerting** with distinct LCD messages and buzzer tones.

4. Hardware Design

4.1 Components

Component	Model / type	Key specification	Application
Microcontroller board	NUCLEO-F401RE (STM32F401RETx, LQFP64)	32-bit ARM Cortex-M4 with FPU, 512 KB flash, 96 KB SRAM	Main processing unit
Temperature sensor	DS18B20	±0.5 °C accuracy from −10 °C to +85 °C; 9–12-bit resolution; 1-Wire	Body temperature
Pulse sensor	Analog optical pulse sensor	Analog output read by 12-bit ADC	Heart rate
Sound sensor	Digital sound-detection module	Digital output pin	Breathing events
Display	16×2 character LCD with I2C adapter	7-bit address 0x27, backlit	Data visualisation

Component	Model / type	Key specification	Application
Buzzer	Piezo/active buzzer	Driven by PWM	Audible alerts
Prototyping	Breadboard and jumper wires in an enclosure	–	Integration

4.2 Pin Assignment

Signal	MCU pin	Mode	Notes
Pulse sensor	PA0	ADC1 channel 0 (analog)	12-bit, single conversion
Buzzer	PA1	TIM2 channel 2 (PWM)	Frequency set at run time
DS18B20 data	PB2	Open-drain output with pull-up / switched to input	Bit-banged 1-Wire
Sound sensor	PB0	Digital input with pull-up	Polled
LCD SCL / SDA	PB6 / PB7	I2C1	100 kHz standard mode
UART TX / RX	PA2 / PA3	USART2	115200 baud via ST-LINK
Status LED / user button	PA5 / PC13	GPIO	On-board LD2 and B1

4.3 Clock and Peripheral Configuration

The 8 MHz external clock (HSE) is divided by 4 (PLLM) and multiplied by 50 (PLLN), then divided by 2 (PLLP), giving a **50 MHz** system clock. AHB and APB2 run at 50 MHz and APB1 at 25 MHz. The configuration was generated with STM32CubeMX and the supporting drivers use the STM32 HAL.

Peripheral	Configuration	Purpose
ADC1	12-bit, right-aligned, software trigger, single conversion, 3-cycle sampling	Pulse sensor (0–4095 counts)
TIM10	Prescaler 49 → 1 MHz tick (1 µs resolution), free-running	Microsecond delays for 1-Wire timing
TIM2 (CH2)	PWM mode 1, prescaler 71, auto-reload preload enabled	Buzzer tone generation
I2C1	100 kHz, 7-bit addressing	LCD communication
USART2	115200 baud, 8 data bits, no parity, 1 stop bit	Serial logging
SysTick	1 ms tick (HAL)	Scheduling and millisecond delays

5. Sensing Methodology

5.1 Body Temperature (DS18B20)

The DS18B20 is a digital temperature sensor that communicates over the Dallas/Maxim **1-Wire** protocol using a single data line plus power and ground. 1-Wire is natively supported by this sensor, keeps wiring minimal and, because the output is digital, avoids signal degradation over cable length. The bus is open-drain: the microcontroller and the sensor can only pull the line low, so a pull-up resistor on the data line returns it to the high level.

The firmware bit-bangs the protocol on PB2, using the 1 µs TIM10 timer for timing:

1. **Reset and presence:** the MCU pulls the line low for 480 µs, releases it and checks (after 80 µs) whether the DS18B20 pulls the line low to signal its presence.
2. **Start conversion:** after a reset, the MCU sends *Skip ROM* (0xCC) and *Convert T* (0x44), then waits 750 ms for the 12-bit conversion.
3. **Read result:** after another reset, the MCU sends *Skip ROM* (0xCC) and *Read Scratchpad* (0xBE) and reads the first two bytes (LSB first).
4. **Conversion:** the 16-bit raw value is divided by 16 to give degrees Celsius (resolution 0.0625 °C).

If no presence pulse is detected, a safe fallback value of 36.5 °C is used so that a disconnected probe does not generate false alarms.

5.2 Heart Rate (Analog Pulse Sensor)

The pulse sensor output is connected to PA0, which is configured as an analog input for ADC1 channel 0. The 12-bit converter produces a value between 0 and 4095:

ADC value = (Vin / Vref) × 4095

In the current firmware, each 5-second cycle takes one ADC conversion and maps the 0–4095 range linearly onto a physiological heart-rate range of 40–100 BPM. Section 13 discusses the planned upgrade to true beat detection (see below), where converted values are used to identify pulse events and BPM is derived from the interval between consecutive beats rather than from the raw amplitude.

5.3 Respiratory Rate (Sound Sensor)

The sound sensor's digital output is connected to PB0, configured as a GPIO input with an internal pull-up. Breathing-related sound events are counted over a **5-second window** (the pin is polled every 100 ms). The count is scaled to breaths per minute:

Respiratory rate (breaths/min) = (events in window ÷ window length in s) × 60 = events × 12

6. Signal Processing

6.1 Exponential Moving Average (EMA) Filter

Raw sensor values fluctuate because of electrical noise, motion and momentary disturbances. The EMA filter smooths this variation while still reacting to every new reading, giving recent samples more weight than older ones:

$EMA(n) = \alpha \cdot x(n) + (1 - \alpha) \cdot EMA(n-1)$, $\alpha = 0.3$

With $\alpha = 0.3$ each new sample contributes 30 % and the history 70 %, a balance between noise rejection and responsiveness. The filter is seeded with the first sample.

Example. For the input sequence 78, 80, 82, 90, 84 (the 90 being a disturbance), the filtered output is 78.0, 78.6, 79.6, 82.7, 83.1. The disturbance is strongly attenuated, yet the filter still tracks the underlying upward movement.

6.2 Circular Buffers

Storing every reading would eventually exhaust RAM, and the analysis only needs recent history. A circular (ring) buffer overwrites the oldest value with the newest once it is full, so memory use is constant. Each parameter has its own buffer of **36 samples**, which is 3 minutes of data at 5-second intervals (180 s ÷ 5 s = 36). A buffer is a small structure holding the data array, a head index and a sample count, with helper functions to push a value, read the n-th oldest value, and compute a windowed mean and standard deviation.

6.3 Multi-Timeframe Analysis

Interval	Operations	Outputs
Every 5 s	Read sensors, push to buffers, update EMA, refresh LCD, log over UART	Filtered current value
Every 15 s	Average of the last 3 samples; rate of change; trend direction	Short-term trend (UP / DN / OK)
Every 1 min	Average of the last 12 samples; mean and standard deviation of the whole buffer; pattern detection; health score; alert	Baseline statistics, score, alert level

The **rate of change** is expressed per minute: the difference between the current filtered value and the value three samples (15 s) earlier, divided by 0.25 min. A trend is classed as *stable* when the absolute rate is below 0.1 unit/min; otherwise it is *rising* or *falling* according to its sign.

7. Pattern Recognition and Anomaly Detection

Pattern detection runs once per minute on each parameter and needs at least five buffered samples. It returns one of four codes: none, spike, drop or oscillation.

- **Spike / drop:** the mean of the last three samples is compared with the mean of the whole buffer (the baseline). If their absolute difference exceeds the parameter's threshold, a *spike* (recent mean higher) or *drop* (recent mean lower) is reported.
- **Oscillation:** if no spike or drop is found and at least six samples exist, the recent samples are scanned for alternating up/down behaviour (local maxima and minima). Two or more reversals are reported as an oscillating pattern, which may indicate physiological instability or an unreliable signal.

Parameter	Spike/drop threshold	Rate-of-change warning level
Temperature	1.0 °C	0.5 °C per minute
Heart rate	20 BPM	10 BPM per minute
Respiratory rate	5 breaths/min	3 breaths/min per minute

8. Health Scoring Algorithm

8.1 Gaussian Normalisation

Each parameter is compared with an optimal value (μ) using a Gaussian-shaped deviation measure. The risk is 0 at the optimum and approaches 1 as the value moves far from it:

$$\text{Risk}(x) = 1 - \exp\left(-(x-\mu)^2 / (2\sigma^2)\right)$$

Parameter	Optimal value (μ)	Spread (σ)
Body temperature	37.0 °C	1.5 °C
Heart rate	70 BPM	15 BPM
Respiratory rate	15 breaths/min	5 breaths/min

Because the curve is Gaussian, small deviations are penalised gently and large deviations saturate smoothly: a reading 1σ from optimum gives a risk of 0.39, 2σ gives 0.86 and 3σ gives 0.99.

8.2 Rate and Pattern Risk

- **Rate risk** for each parameter is the absolute rate of change divided by its warning level, limited to the range 0–1.
- **Pattern risk** adds 0.2 if temperature shows a pattern, 0.3 for heart rate and 0.2 for respiration, limited to a maximum of 0.5.

8.3 Composite Score

The three components are fused with fixed weights: base deviation 60 %, rate of change 30 % and pattern 10 %. Within the base and rate terms the parameters are weighted 0.2 (temperature), 0.5 (heart rate) and 0.3 (respiration):

$R_{total} = 0.6 \times (0.2 \cdot R_T + 0.5 \cdot R_{HR} + 0.3 \cdot R_{RR}) + 0.3 \times (0.2 \cdot r_T + 0.5 \cdot r_{HR} + 0.3 \cdot r_{RR}) + 0.1 \times P$

Health score $H = (1 - R_{total}) \times 100$, limited to 0–100

Higher scores indicate better condition. Since the largest possible R_{total} is 0.95, the lowest score the algorithm can produce is 5.

Worked example (base term only). For a filtered temperature of 34.1 °C, heart rate of 85 BPM and respiratory rate of 5/min, the individual risks are 0.85, 0.39 and 0.86. The base term is $0.6 \times (0.2 \times 0.85 + 0.5 \times 0.39 + 0.3 \times 0.86) \approx 0.375$, which on its own would remove about 37.5 points from the score before the rate and pattern terms are added.

9. Alert Management

The health score is mapped to one of five alert levels. Each level has its own LCD message and buzzer pattern, and is also reported over UART.

Score	Level	LCD message	Buzzer
⩾ 70	0 – None	HEALTH NORMAL	Silent
55 – 69.9	1 – Low	MINOR CONCERN / Monitor closely	1 beep at 1000 Hz, 100 ms
40 – 54.9	2 – Medium	CAUTION REQD / Check vitals	2 beeps at 1500 Hz, 150 ms
25 – 39.9	3 – High	WARNING!!! / Seek attention	3 beeps at 2000 Hz, 100 ms
< 25	4 – Critical	CRITICAL!!! / Immediate help	5 short beeps at 2500 Hz, 80 ms

Buzzer tones are generated by changing the TIM2 auto-reload value to set the PWM frequency, with a 50 % duty cycle.

10. Communication Interfaces

Protocol	Application	Configuration
1-Wire	DS18B20 temperature sensor	Single data line (plus VDD and GND), open-drain with pull-up
ADC	Pulse sensor	12-bit, single conversion
GPIO input	Sound sensor	Digital level, polled every 100 ms
I2C	LCD	100 kHz standard mode, address 0x27
UART	Data logging to PC	115200 baud, 8-N-1, ST-LINK virtual COM port
PWM	Buzzer	Variable frequency, 50 % duty

I2C. I2C is a synchronous, serial, half-duplex bus: the SDA line carries both address and data, while SCL provides the clock. A transfer consists of a START condition, the 7-bit slave address with the read/write bit, an acknowledge (ACK) from the slave, one or more data bytes each followed by an ACK, and a STOP condition. The MCU sends the formatted text (temperature, heart rate, respiratory rate, health score) to the LCD this way.

UART. The same information is sent to a PC through the ST-LINK virtual COM port and viewed in a terminal program such as PuTTY, providing a timestamp-free but complete record of every 5-second sample, 15-second trend update and 1-minute analysis.

11. Firmware Design

11.1 Development Tools

Tool	Version	Purpose
STM32CubeMX	6.15.0	Pin, clock and peripheral configuration
STM32CubeIDE	1.19.0	Editing, building and debugging
STM32Cube FW_F4 (HAL)	V1.28.3	Hardware abstraction layer
I2C LCD driver	Custom	Character LCD interface (init, clear, set cursor, send string)

Tool	Version	Purpose
PuTTY	–	Serial log viewing

11.2 Software Modules

Module	Key functions	Role
Circular buffer	CB_Init, CB_Push, CB_Get, CB_Average, CB_StdDev	Time-series storage and statistics
EMA filter	EMA_Update	Noise reduction
Trending	Trending_Update5s, Trending_Analyze15s, Trending_Analyze1min	Filtered value, short and long-term analysis
Pattern detection	Detect_Pattern	Spike, drop, oscillation
Scoring	norm_temp_0_1, norm_bpm_0_1, norm_resp_0_1, compute_health_score_with_trends	Gaussian risk and composite score
Alerts	Determine_Alert_Level, Handle_Alert	Level mapping, LCD, buzzer, UART
Sensors	DS18B20_Start / Write / Read, adc_read_raw, Update_HeartRate, Update_Breath	Acquisition
Buzzer	Buzzer_SetFrequency, Buzzer_Beep	PWM tones

Three data structures hold the state: a **circular buffer** (36-float array, head index, count), a **trending record** per parameter (current EMA value, short and long averages, rate of change, baseline mean and standard deviation, trend direction, pattern code) and an **alert-level enumeration** (none to critical).

11.3 Initialisation Sequence

1. HAL initialisation and system clock configuration (50 MHz from the PLL).
2. GPIO, USART2, I2C1, TIM10, ADC1 and TIM2 initialisation.
3. Start the TIM10 microsecond time base.
4. Initialise the LCD and show the start-up and “Initializing Buffers” screens.

5. Clear the three circular buffers.
6. Send a start-up banner over UART and record the start times of the 5 s, 15 s and 1 min schedulers.

11.4 Main Loop

The main loop is a simple time-based scheduler using the millisecond HAL tick. On every pass it checks three elapsed-time conditions and runs the corresponding stage:

- **5 s:** read DS18B20, take one ADC conversion, count breathing events over the 5-second window, push all three values into their buffers, update the EMAs, show temperature, heart rate and breathing rate on the LCD and log them over UART.
- **15 s:** compute short-term averages, rate of change and trend direction; show the trend and heart-rate slope on the LCD; log the trend line.
- **1 min:** compute long-term statistics, run pattern detection, compute the health score and alert level, display pattern warnings, statistics and the score, send the full analysis report over UART and finally trigger the alert response.

11.5 Resource Usage

Memory use is fixed at build time. The three circular buffers and three trending records together occupy roughly half a kilobyte of RAM, out of 96 KB available. The firmware itself is small (on the order of tens of kilobytes of the 512 KB flash; the exact figure is given by the build output). Floating-point arithmetic, including `expf` and `sqrtf`, runs on the Cortex-M4's single-precision FPU.

12. Results and Analysis

12.1 Test Setup

The prototype was assembled on a breadboard inside an enclosure, with the LCD mounted in a front-panel cut-out and the temperature probe, pulse sensor and sound sensor connected externally. Output was observed on the LCD and on a PC terminal connected at 115200 baud.

12.2 Sample Serial Log

A captured session shows the three reporting levels working together:

```
[5s] T:34.1 HR:87 R:1
[5s] T:34.1 HR:85 R:5
[15s] Trends - T:DN(-0.26) HR:UP(0.6) R:UP(18.0)

[1min] === COMPREHENSIVE ANALYSIS ===
Temp: 34.1C (mean:33.8 std:0.54)
HR: 85 bpm (mean:93 std:8.1)
Resp: 5/min (mean:5.5 std:10.1)
Health Score: 48.6/100
Alert Level: 2
==================================

ALERT: Caution Required
```

12.3 Interpretation

- The 5-second lines show the filtered values being refreshed regularly.
- The 15-second line reports direction and rate for each parameter (for example, temperature falling at 0.26 °C/min).
- The 1-minute report gives the rolling baseline (mean and standard deviation), the composite score of 48.6/100 and alert level 2, which falls in the 40–55 band and correctly triggers the *Caution Required* alert on the LCD, buzzer and UART.
- The score is lowered mainly by the low skin-contact temperature and low breathing count relative to their optima, together with the large rate-of-change values caused by variable breathing counts. The large respiratory standard deviation (10.1) reflects this variability and shows how the baseline statistics expose unstable signals.

12.4 Performance Summary

Metric	Specification / design value	Implementation
Temperature accuracy	±0.5 °C (DS18B20 datasheet, −10 to +85 °C)	12-bit reading, 0.0625 °C resolution
Heart-rate range	40–100 BPM	Linear ADC mapping
Respiratory rate	Breaths per minute from a 5-second count	Resolution of 12 breaths/min per counted event
Sampling interval	5 seconds	Nominal 5 s cycle

Metric	Specification / design value	Implementation
Trend / analysis updates	15 s and 1 min	Scheduled from the SysTick timer
Alert generation	Within the 1-minute analysis cycle	LCD, buzzer and UART together
Memory	Fixed allocation	36 samples × 3 parameters

13. Limitations and Future Scope

- **True heart-rate detection:** replace the linear ADC mapping with beat detection. Pulse events are identified from the waveform (adaptive threshold or peak detection), the time of each beat is recorded, and BPM = 60 000 ms ÷ inter-beat interval, with the ADC read at a fixed rate from a timer interrupt or DMA. This removes the 40–100 BPM ceiling and gives physiologically meaningful values.
- **Respiration sensing:** a sound-event count over 5 seconds is coarse (each event equals 12 breaths/min). A longer averaging window, an envelope detector, or a dedicated airflow, piezo or thermistor sensor would improve accuracy.
- **Skin versus core temperature:** surface readings are lower than core body temperature. A calibration offset or a better-coupled probe would bring readings closer to the 37 °C optimum used by the scoring model.
- **Non-blocking design:** the sensor conversion wait, breathing window and alert delays use blocking delays, so the true cycle is slightly longer than 5 s and the system is unresponsive during these periods. A timer-interrupt or RTOS-based design would remove this.
- **Buzzer tone accuracy:** the tone-frequency calculation should be derived from the actual timer clock so that produced frequencies match the nominal values.
- **Connectivity and storage:** add BLE or Wi-Fi telemetry, a mobile or web dashboard, and non-volatile logging (SD card or flash) for long-term records.
- **Personalised thresholds:** adaptive baselines per user, and machine-learning classification of patterns, could replace fixed thresholds.
- **Validation:** clinical-grade validation and comparison with reference instruments are required before any medical use.

14. Conclusion

The project demonstrates a complete embedded health-monitoring chain on an STM32 microcontroller: multi-sensor acquisition, EMA noise filtering, constant-memory circular buffering, multi-timeframe trend analysis, statistical pattern detection, Gaussian risk scoring and a five-level alert system with LCD, buzzer and UART outputs. By combining current values with their rate of change and recent patterns, the system provides earlier and more informative warnings than single-threshold alarms, and its fixed memory footprint and modest computation make it well suited to low-power, low-cost wearable and remote-monitoring devices. The identified improvements, chiefly beat-based heart-rate detection, a better respiration sensor and a non-blocking firmware structure, define a clear path towards a more accurate and deployable system.

Appendix A: Configuration Constants

Constant	Value	Meaning
BUFFER_SIZE	36	Samples per circular buffer (3 min at 5 s)
EMA_ALPHA	0.3	EMA smoothing factor
SHORT_WINDOW	3	Samples for 15 s analysis
LONG_WINDOW	12	Samples for 1 min analysis
OPTIMAL_TEMP_C / TEMP_SPREAD	37.0 / 1.5	Gaussian parameters, temperature
OPTIMAL_BPM / BPM_SPREAD	70 / 15	Gaussian parameters, heart rate
OPTIMAL_RESP / RESP_SPREAD	15 / 5	Gaussian parameters, respiration
TEMP / BPM / RESP_RATE_WARN	0.5 / 10 / 3	Rate-of-change warning levels (per minute)
TEMP / BPM / RESP_SPIKE_THRESH	1.0 / 20 / 5	Spike and drop thresholds

Appendix B: Abbreviations

Abbreviation	Full form
ADC	Analog-to-Digital Converter
ARM	Advanced RISC Machines
BPM	Beats Per Minute
EMA	Exponential Moving Average
FPU	Floating-Point Unit
GPIO	General-Purpose Input/Output
HAL	Hardware Abstraction Layer
I2C	Inter-Integrated Circuit
LCD	Liquid Crystal Display
MCU	Microcontroller Unit
PWM	Pulse Width Modulation
STM32	STMicroelectronics 32-bit microcontroller family
UART	Universal Asynchronous Receiver-Transmitter

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